

GROUP MEETING

ALIGNNN and the PINK Pipeline

A second architecture, the full GNoME screen, validation, an audit, and a 15-crystal DFT list



What PINK does, in one slide

PINK (Liu *et al.*, *J. Mater. Inf.* 2025) predicts **lattice thermal conductivity** κ_L from a CIF file, without solving the phonon Boltzmann transport equation.

A graph network predicts two elastic moduli; a closed-form model turns them into κ_L . That physics stage has **no parameters fitted to this data** (slide 16).

Why bother

DFT + phonons take days per material; this takes about a sixth of a second – enough to screen half a million.

The network never sees a thermal conductivity. It learns elasticity; the physics does the rest.

The two stages:

Stage 1 – learned

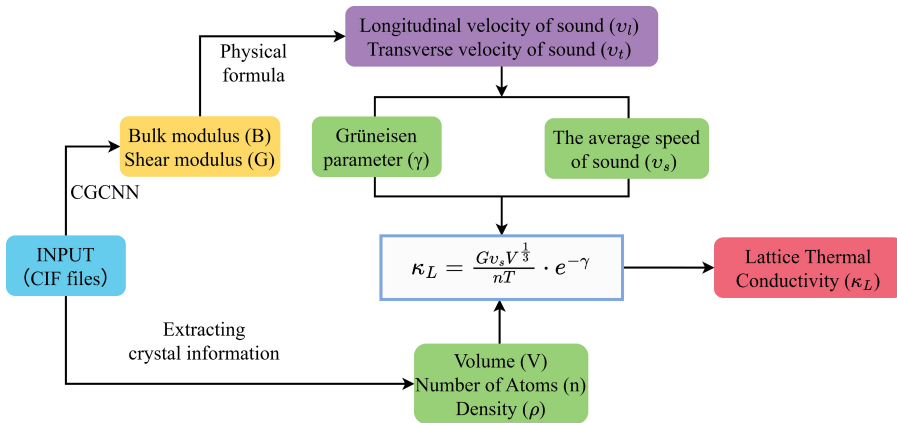
CIF $\rightarrow$ graph network $\rightarrow$ bulk modulus K , shear modulus G



Stage 2 – physics

$K, G \rightarrow$ Poisson ratio $\rightarrow$ sound velocity, Debye temperature, Grüneisen $\gamma \rightarrow \kappa_L$

The paper's own Figure 1



Reproduced from Liu *et al.*, *J. Mater. Inf.* 2025, 5, 12 (CC BY 4.0). **Blue** the input CIF; **yellow** the only learned step; **purple and green** the closed-form physics; **red** the answer. The boxed equation is the one this project reports – note $\kappa_L \propto e^{-\gamma}$, an *exponential* dependence on the Grüneisen parameter.

What this project added

A second architecture

6-14

ALIGNN, trained on the identical split, so the two compare directly.

The full GNoME screen, twice

21-24

33,053 candidates scored by both architectures.

Checks against a real answer

18-20

45 measured κ_L values and phonon-DFT κ_L for thousands of crystals, not only model vs model.

An audit

25-26

Every stored metric recomputed from its inputs: 9,048 values, 0 mismatches.

A list for phonon DFT

27-30

15 crystals, 4 high- κ controls and 8 reserves, handed over for a DFT κ_L .

Two architectures agreeing is consistent with a physical signal, not proof of one. The measured and phonon-DFT checks are what test it.

The models in this deck

name used here	what it is	slides
CGCNN $\times 1 / \times 3$	graph network on bond lengths. $\times 1$ = seed 42; $\times 3$ = seeds 42, 1, 2, averaged in $\log_{10}$	10-14
ALIGNN $\times 1 / \times 3$	graph network that also sees bond angles; same seeds and averaging	6-14
descriptor XGBoost	boosted trees on 297 matminer descriptors; the 3rd moduli model	27-29
composition-only trees	random forest and XGBoost on the formula alone: the baselines	26
direct ALIGNN	structure $\rightarrow \kappa_L$ in one step, trained on phonon-DFT κ_L ; no Slack chain	20, 28-29
Poisson γ	Grüneisen γ from the predicted K/G (Eq. 6): the paper's route	15-20
MLIP γ	γ from phonons computed with a machine-learned potential	27-30

All moduli models: same matbench training set, same test split (slide 10).

Two kinds of model: those that predict K , G and need the physics, and one (direct ALIGNN) that skips it.

Why a second architecture at all

A single model that scores well can be right for the wrong reason. Two models that share the training data and the physics, but not the architecture, give a cross-check: where they agree, the result is less likely to be an artefact of either one.

ALIGNN was chosen because it differs from CGCNN in a **physically motivated** way. It is not a bigger CGCNN: it sees something CGCNN cannot – **bond angles**.

What CGCNN encodes

Which atoms exist, which are bonded, and *how far apart* they are.
A two-body description.

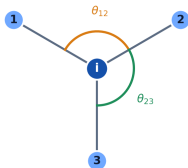
What ALIGNN adds

The *angles* between bonds – a three-body term. Resistance to shear depends on how bonds resist *bending*, not only stretching.

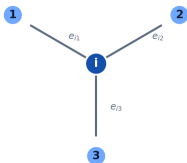
Shear modulus is a bond-bending property. A model blind to angles has to infer it indirectly.

One crystal becomes *two* graphs

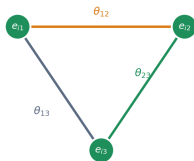
A the crystal
atoms, bonds, and the angles between them



B the ATOM graph $\mathcal{G}$
nodes = atoms, edges = bonds (CGCNN stops here)



C the LINE graph $\mathcal{L}$
nodes = bonds, edges = ANGLES (ALIGNN adds this)



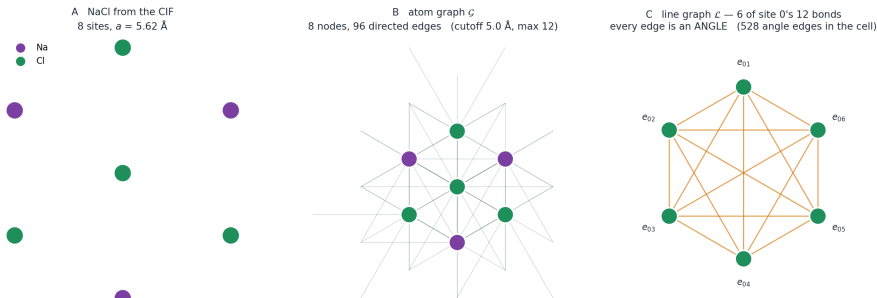
Neighbours. A periodic search from the CIF: within **5.0 Å**, at most **12** per atom (the trained model's config). Atoms start from CGCNN's 92-number element table.

B: atom graph $\mathcal{G}$. Atoms are nodes, bonds are edges carrying the length r_{ij} . This is what CGCNN builds, and where it stops.

C: line graph $\mathcal{L}$. Every *bond* becomes a node; two bonds sharing an atom are joined by an edge carrying their angle, fed as $\cos \theta_{jik}$ on a radial basis.

An angle belongs to a *pair* of bonds, so it needs a graph whose nodes are bonds. That graph is $\mathcal{L}$.

The same thing on a real file: NaCl



Computed from `nacl.cif` at **ALIGNN's own** settings – cutoff $5.0 \text{ \AA}$, max 12 neighbours, from the trained model's config.

A **bond is two things at once**: an edge of $\mathcal{G}$, a node of $\mathcal{L}$.

	node	edge	edge feature
$\mathcal{G}$	atom (8)	bond (96)	length r_{ij}
$\mathcal{L}$	bond (96)	angle (528)	angle θ_{jik}

12 nbrs $\rightarrow \binom{12}{2} = 66$ angles, $8 \times 66 = 528$, so

$528/96 = 5.5\times$ the edges. Panel **C** shows site 0 only.

One ALIGNN layer, two graphs

One ALIGNN layer is two coupled updates, run in order:

1. Update on the line graph $\mathcal{L}$

Each bond-node is updated from the bonds it shares an atom with, using the *angle* on the connecting edge. Bond features now carry angular information.



2. Update on the atom graph $\mathcal{G}$

Each atom-node is updated from its neighbours using those *angle-aware* bond features as edge inputs.

Then, as in CGCNN

Average the atom vectors over the cell, pass them through a small fully-connected head, and emit one number.

Trained on GPa, scored in $\log_{10}$

ALIGNN is trained on the modulus itself, in GPa, with a mean-squared-error loss; its $\log_{10}$ errors are computed afterwards (predictions are floored at 10^{-3} GPa). CGCNN trains on $\log_{10}$ directly. Both train one network per modulus.

CGCNN and ALIGNN, side by side

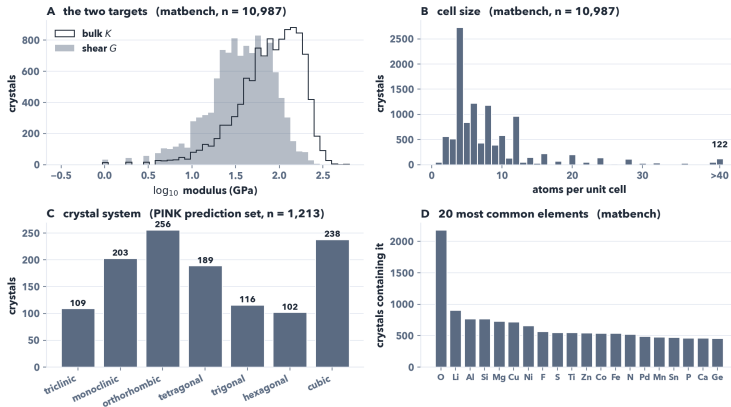
	CGCNN	ALIGNN
graphs built	atom graph only	atom graph + line graph
geometry seen	bond lengths (two-body)	lengths + angles (three-body)
neighbours	12 nearest within 8 Å	12 nearest, 5 Å cutoff
layers, width	3 graph convolutions*, 64 wide	4 ALIGNN + 4 GCN, 256 wide
trainable weights	80,833*	≈4.0 million
training target	$\log_{10} K$ or $\log_{10} G$	K or G in GPa
optimiser, epochs	Adam, lr 0.01, 200	AdamW, lr 0.001, 150
bulk K MAE, $\log_{10}$: $\times 1 / \times 3$	0.0696 / 0.0630	0.0549 / 0.0534
shear G MAE, $\log_{10}$: $\times 1 / \times 3$	0.0836 / 0.0781	0.0726 / 0.0708

Same 10,987 crystals, same 7,691 / 1,648 / 1,648 split (`split_seed = 42`). $\times 1$ = seed 42; $\times 3$ = seeds 42, 1, 2 averaged in $\log_{10}$. Both: mean-squared-error loss, weight decay 10^{-5} . *CGCNN seed 1 has 4 convolutions (102,977 weights), so the CGCNN ensemble mixes two depths.

One ALIGNN beats the CGCNN ensemble on both moduli. The recipes differ too: credit architecture *plus* recipe.

11 • ACCURACY

The training data



A: both targets span more than two orders of magnitude, so errors are measured in $\log_{10}$. G sits below K and reaches lower.

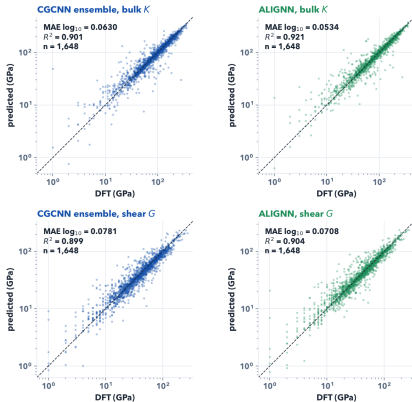
B: the median cell holds 6 atoms; 122 hold more than 40.

C is a *different* set: the 1,213 crystals the paper predicts for (slide 19). One CIF (mp-770620) has two atoms 0.0009 Å apart, so its crystal system is read from the CIF header – all 1,213 are shown.

D: 20% of crystals contain oxygen.

matbench: 10,987 DFT elastic tensors (Materials Project). Oxides dominate; soft crystals are rare.

Predicted against DFT, both ensembles



The 1,648 test crystals, never seen in training. Each point is one crystal; the dashed line is a perfect prediction.

R^2 on $\log_{10}$, **3-model ensembles**

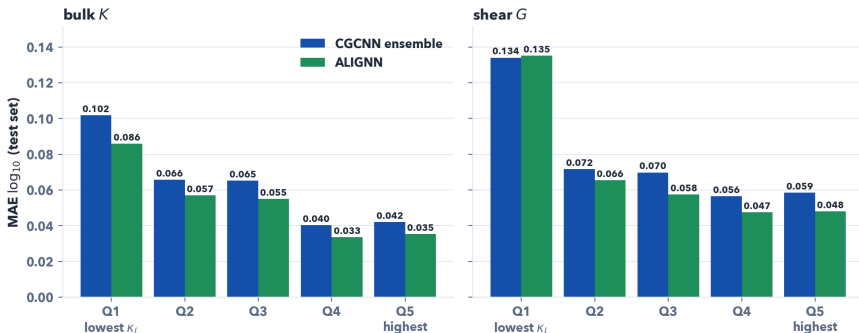
CGCNN K 0.901 G 0.899

ALIGNN K 0.921 G 0.904

Every test prediction is positive (lowest: ALIGNN K, 0.623 GPa). The negative K in the earlier version of this figure was a validation-set prediction of a model no longer used.

Both fit well across the range. The scatter widens at the soft end – where the screen looks.

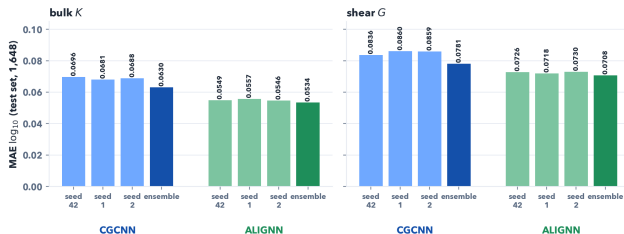
Where the models fail: the soft end



Test crystals in five equal bins (330 each) by κ_L from the DFT moduli; Q1 = the lowest fifth, the only bin a low- κ screen reads. Shear error in Q1 is $2.3\times$ that in Q5 for CGCNN (0.134 vs 0.059) and $2.8\times$ for ALIGNN (0.135 vs 0.048).

ALIGNN is better in every bin but one: shear modulus in Q1, where the two tie – exactly where the screen looks.

What ensembling buys



MAE cut by averaging three models	vs seed 42	vs mean member	95% CI of the change vs seed 42
CGCNN K	9.4%	8.4%	[-0.0085, -0.0046]
CGCNN G	6.5%	8.3%	[-0.0075, -0.0035]
ALIGNN K	2.7%	3.1%	[-0.0024, -0.0006]
ALIGNN G	2.6%	2.4%	[-0.0030, -0.0007]

CI: paired bootstrap of (ensemble MAE – seed-42 MAE) in $\log_{10}$ over the 1,648 test crystals. All four lie below zero. CGCNN seed 1 is one layer deeper (slide 10).

Averaging helps CGCNN about three times as much as ALIGNN; one ALIGNN still beats the CGCNN ensemble.

15 • PHYSICS

From K , G to sound speed, Θ_D and γ

$$v_l = \sqrt{\frac{K + \frac{4}{3}G}{\rho}} \quad (1)$$

$$v_t = \sqrt{\frac{G}{\rho}} \quad (2)$$

$$v_s = \left[\frac{1}{3} \left(\frac{1}{v_l^3} + \frac{2}{v_t^3} \right) \right]^{-1/3} \quad (3)$$

$$\Theta_D = \frac{h}{k_B} \left(\frac{3}{4\pi V} \right)^{1/3} v_s \quad (4)$$

$$\nu = \frac{(v_l/v_t)^2 - 2}{2(v_l/v_t)^2 - 2} \quad (5)$$

$$\gamma = \frac{3(1 + \nu)}{2(2 - 3\nu)} \quad (6)$$

K , G bulk and shear modulus (*predicted*, GPa) · ρ density (CIF) · v_l , v_t , v_s longitudinal, transverse, average sound speed · ν Poisson's ratio · γ Grüneisen parameter · V volume of the CIF's *primitive* cell, as in the paper's released code (the textbook Debye formula uses volume per atom).

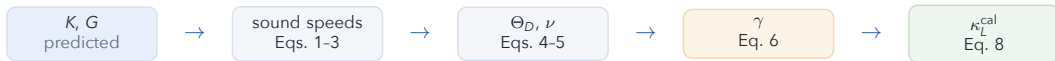
Eq. 6 matters most: γ is *derived* from the same K , G – the “Poisson γ ”.

The κ_L formula, and “the Slack chain”

$$\kappa_L^{\text{Slack}} = \frac{A(\gamma) \bar{M} V^{1/3} \Theta_D^3}{\gamma^2 T N} \quad \text{Slack's model, } A(\gamma) = \frac{2.43 \times 10^{-8}}{1 - 0.514/\gamma + 0.228/\gamma^2} \quad (7)$$

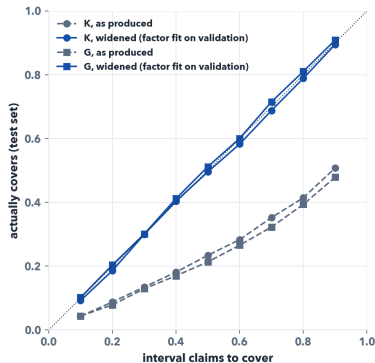
$$\kappa_L^{\text{cal}} = \frac{G v_s V^{1/3}}{N T} e^{-\gamma} \quad \text{the boxed equation of the paper's Figure 1} \quad (8)$$

$\bar{M}$ mean atomic mass (amu; the released code puts in the total cell mass), N atoms per primitive cell, $T = 300$ K. No number here uses Eq. 7. **This project reports κ_L^{cal} :** changing γ rescales it by $e^{\gamma_{\text{old}} - \gamma_{\text{new}}}$, which is how the MLIP γ is swapped in later.



In this deck, “the Slack chain” means this path. Only K and G are learned; no parameter of the chain is fitted to this data.

The 90% interval was overconfident



The p05-p95 band for K and G from the three CGCNN members' spread, checked against the test split's DFT labels.

Claims 90%. Covers 51% (K), 48% (G).

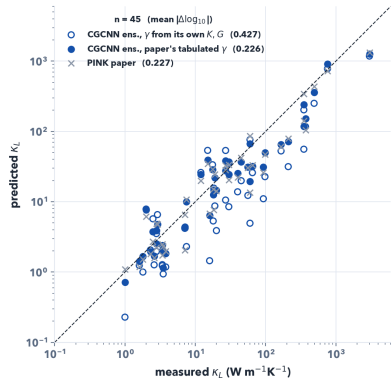
Identically trained members share blind spots, so they disagree less than they err.

Widened, it is honest

Widening by $4.27\times$ (K) and $4.54\times$ (G), fitted on the validation split, gives 90% and 91% on the test split.

The members' spread cannot see the error that all three share.

Measured κ_L , on equal footing



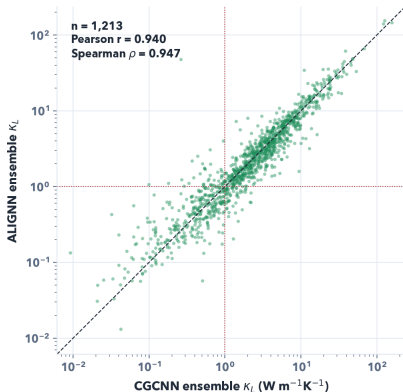
PINK's Table 1: 45 materials with **measured** κ_L (the 46th, GaP, is deprecated in Materials Project); 43 of them are in our moduli training set. **Paper's** γ = the paper's tabulated values, from databases and experiment, not Eq. 6.

	mean $ \Delta \log_{10} $	own γ (Eq. 6)	paper's γ
CGCNN ensemble	0.427	0.226	
ALIGNN ensemble	0.450	0.246	
PINK paper			0.227

κ_L , W/m/K	measured	CGCNN ensemble		PINK
		own γ	paper's γ	
AgCl	1.0	0.23	0.71	1.09
PbTe	2.5	5.72	3.74	2.71
C (diamond)	3000.0	1180.18	1260.63	1320.87

With the paper's γ we match the paper (0.226 vs 0.227); with Eq. 6's γ the error doubles. γ , not the moduli, is the weak link.

ALIGNN vs CGCNN on PINK's crystals



Both ensembles' moduli through the same Slack chain, on the 1,213 crystals the paper predicts for.

Agreement, ALIGNN vs CGCNN

κ_L $r = 0.940$, $\rho = 0.947$

MAE $0.141 \log_{10}$

K alone $r = 0.961$

G alone $r = 0.966$

The moduli agree more closely than κ_L does: the chain, through γ , amplifies small K/G differences.

Consistent with a physical signal – but both share the same γ route, so this agreement cannot test γ .

Against phonon DFT

test set	model	measured as	value
PhoNIX, not in matbench 2,520 crystals, ≤ 20 atoms	Slack + Poisson γ , highest of 3 each model alone highest of 3 a blind pick	precision at $\kappa_L \leq 1$	0.747 [0.704, 0.785]
		precision at $\kappa_L \leq 1$	0.639-0.650
		recall (true lows found)	0.581
		base rate (truly ≤ 1)	0.225
PhoNIX, 5-fold 5,316 held out once	direct ALIGNN, structure $\rightarrow \kappa_L$	MAE $\log_{10}$	0.194
		R^2 / Spearman ρ	0.755 / 0.892

The Slack chain, blind

PhoNIX is a published set of phonon-DFT κ_L . Of the crystals our chain calls ≤ 1 , **3 in 4** really are, against 0.225 for a blind pick. Taking the highest of three moduli models raises precision from ≈ 0.64 to 0.747 – the reason the DFT list uses that rule.

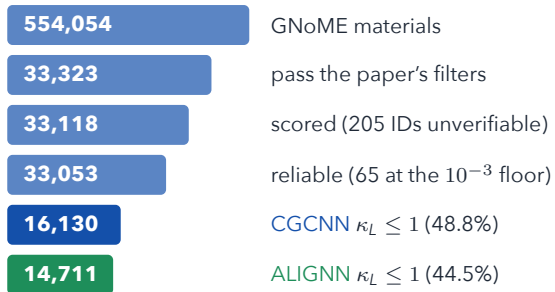
Direct ALIGNN

Trained straight on phonon-DFT κ_L , no Slack formula. Each crystal is predicted once, while held out: average miss $\times 1.56$ ($10^{0.194}$), rank $\rho = 0.892$.

This is the closest thing to ground truth so far: the chain's low calls are right about three times in four.

21 • THE SCREEN

The screen, rerun (PINK's Figure 7)



Against the paper's own list

The paper published 11,869 candidates. Of those,

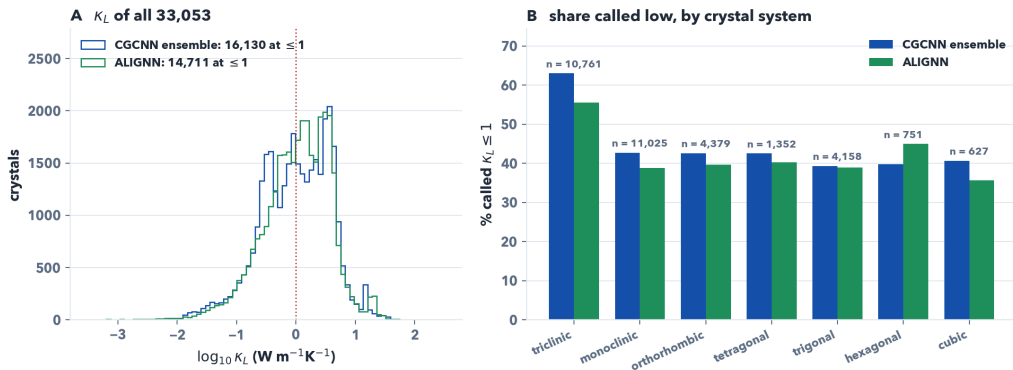
CGCNN also calls8,327 (70.2%)

ALIGNN also calls7,752 (65.3%)

Today's GNoME holds 554,054 materials against the paper's 377,221, so the two lists are not directly comparable.

One count chain for the whole deck: 554,054 → 33,323 → 33,118 → 33,053.

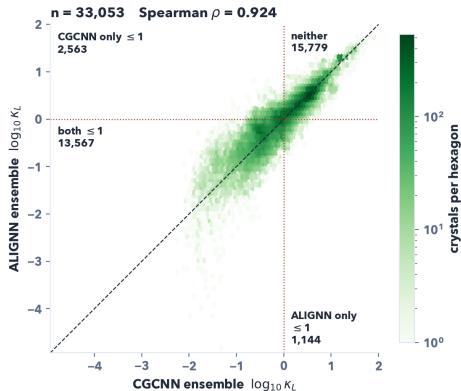
The candidates (PINK's Figure 8)



A: predicted κ_L for all 33,053 reliable candidates; the dotted line is the $\kappa_L = 1$ cut. **B:** the share called low within each crystal system (n above each pair).

The models call 44.5% (ALIGNN) and 48.8% (CGCNN) low; triclinic crystals are the most enriched under both.

ALIGNNN vs CGCNN, the whole screen



The same 33,053 candidates, one point per crystal (binned into hexagons). Only the moduli differ; the Slack chain is identical.

The $\kappa_L \leq 1$ call

both low	13,567
CGCNN only	2,563
ALIGNNN only	1,144
neither	15,779
Spearman ρ	0.924

The two agree on ranking ($\rho = 0.924$); where they split, CGCNN is the more generous by about two to one.

Which elements make low- κ candidates

element	candidates	enrichment	
		CGCNN	ALIGNN
Br	3,281	1.88	2.12
I	3,185	1.86	1.98
Cl	1,331	1.78	1.84
Cs	3,613	1.73	1.86
Rb	3,844	1.66	1.83
Mo	1,792	0.35	0.40
N	1,224	0.40	0.35
W	1,352	0.28	0.30
O	11,538	0.27	0.30
Gd	772	0.18	0.16

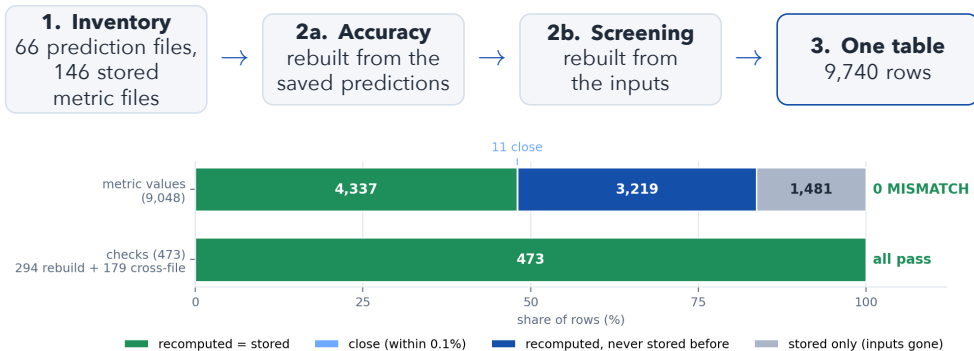
Enrichment = the share of candidates containing the element that a model calls low, divided by that model's share over all candidates. 1 = no effect; 2 = twice as likely to be called low.

Top five and bottom five of the 73 elements present in at least 400 candidates.

Heavy, soft halides and the large alkali metals sit high; O, N and the refractory metals sit low. That is what Eqs. 1-3 predict: a low G gives a low sound speed.

The two models agree on which elements matter. They split on fluorine (1.63 vs 0.90), not shown in the table.

Every metric rebuilt: 0 mismatches



9,740 rows = 9,048 metric values + 473 checks (294 rebuild, 179 cross-file) + 219 repeats of a value stored in two files. **match**: rebuilt, identical. **close**: within 0.1% (GPU models re-run on a CPU). **new**: never stored before. **stored only**: inputs gone, not rebuilt.

9,048 values, 0 MISMATCH; all 473 checks pass. Formulas were re-written, not imported, so an old bug could not hide.

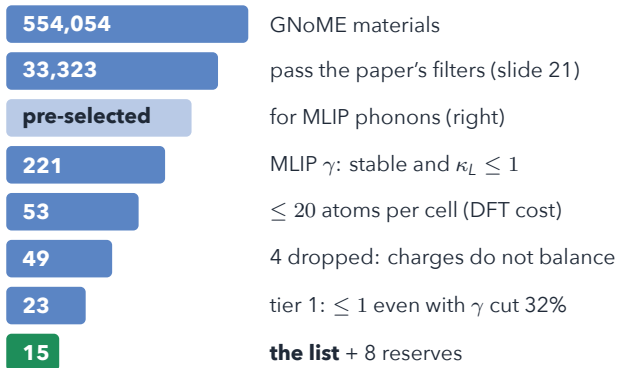
The moduli models, from that one file

model	bulk K		shear G		κ_L
	MAE	R^2	MAE	R^2	MAE
Random forest (composition only)	0.0868	0.841	0.1062	0.830	–
XGBoost (composition only)	0.0879	0.846	0.1128	0.820	–
CGCNN, 1 model (seed 42)	0.0696	0.882	0.0836	0.889	0.2065
CGCNN, 3-ensemble	0.0630	0.901	0.0781	0.899	0.1897
ALIGNN, 1 model (seed 42)	0.0549	0.917	0.0726	0.900	–
ALIGNN, 3-ensemble	0.0534	0.921	0.0708	0.904	0.1708

All on the same 1,648 matbench test crystals (`split_seed = 42`); MAE and R^2 in $\log_{10}$. The κ_L column scores the Slack chain on *predicted* moduli against the same chain on the *DFT* moduli – not against a phonon calculation (that is slide 20). The composition-only trees never had a κ_L run.

ALIGNN's 3-ensemble is best in every column, and both graph networks beat both composition-only trees.

From 554,054 materials to 15 crystals



How crystals reached the MLIP stage

Halides: the top 300 by three moduli models, from a pool pre-ranked by an earlier model set.

Oxides: CGCNN ensemble $\kappa_L \leq 1$.

Ranked by safety, not by lowest value

The goal is a DFT κ_L below 1, so the ranking uses κ_{stress} : the *highest* of three moduli models (CGCNN, ALIGNN, descriptor XGBoost), with the MLIP γ cut by 32%.

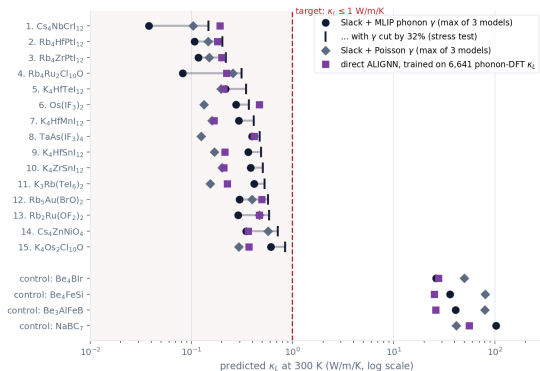
Every stage is a file in the repo (steps 83-91). Three crystals that need rare oxidation states were moved to the reserves.

The 15, as handed to the supervisor

#	formula	atoms	space group	supercell	spin-polarise	predicted κ_L , W/m/K, 300 K			
						max3	stress	Poisson	direct
1	Cs ₄ NbCrI ₁₂	18	P4/mmm (123)	2×2×1 (72)	yes, Cr(III)	0.04	0.15	0.10	0.19
2	Rb ₄ HfPtI ₁₂	18	P4/m (83)	2×2×1 (72)	no	0.11	0.20	0.15	0.18
3	Rb ₄ ZrPtI ₁₂	18	P4/m (83)	2×2×1 (72)	no	0.12	0.22	0.15	0.20
4	Rb ₄ Ru ₂ Cl ₁₀ O	17	I4/mmm (139)	2×2×1 (68)	check, Ru(IV)	0.08	0.31	0.26	0.22
5	K ₄ HfTeI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	no	0.22	0.35	0.20	0.21
6	Os(IF ₃) ₂	18	P $\bar{1}$ (2)	2×2×2 (144)	yes, Os(V) + I ₂ ⁺	0.28	0.37	0.13	0.47
7	K ₄ HfMnI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	yes, Mn(IV)	0.29	0.41	0.16	0.17
8	TaAs(IF ₃) ₄	18	Cm (8)	2×2×2 (144)	check, I ₂ ⁺	0.40	0.47	0.12	0.42
9	K ₄ HfSnI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	no	0.36	0.49	0.17	0.21
10	K ₄ ZrSnI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	no	0.39	0.51	0.20	0.21
11	K ₃ Rb(TeI ₆) ₂	18	P1 (1)	2×2×1 (72)	no	0.42	0.53	0.15	0.23
12	Rb ₅ Au(BrO) ₂	20	Pbam (55)	2×2×1 (80)	no	0.30	0.57	0.40	0.50
13	Rb ₂ Ru(OF ₂) ₂	18	Ama2 (40)	2×2×2 (144)	yes, Ru(VI)	0.29	0.59	0.47	0.47
14	Cs ₄ ZnNiO ₄	10	C2/m (12)	3×2×2 (120)	yes, Ni(II)	0.35	0.72	0.57	0.36
15	K ₄ Os ₂ Cl ₁₀ O	17	I4/mmm (139)	2×2×1 (68)	check, Os(IV)	0.61	0.85	0.30	0.37

max3: Slack chain with MLIP γ , highest of 3 moduli models. **stress:** max3 with γ cut 32% (the ranking value). **Poisson:** Slack chain with Poisson γ , highest of 3. **direct:** direct ALIGNN. Supercell: smallest with every vector ≥ 10 Å. 8 of 15 are one A₂BI₆-type iodide family; a 2026 DFT study of 18 Cs₂BX₆ found every $\kappa_L < 1$ (Cao, arXiv:2602.06501). None of the 15 has a lab-measured (ICSD) structure in Materials Project, and a by-hand formula search (2026-10-06) found no report of any of them; 12 have a reported parent family or anion (step 90).

Three routes, and four controls



Navy circle: Slack chain with MLIP γ (max3); the bar runs to the stress value.

Slate diamond: Slack chain with Poisson γ – shares the moduli, not the γ .

Purple square: direct ALIGNN – shares neither.

Highest value of any route on the 15: **0.85**

Lowest on the 4 controls: **25.3**

Poisson route, stricter 0.54 cut: **14 of 15** ($\text{Cs}_4\text{ZnNiO}_4$ 0.57)

Direct route: **15 of 15** ≤ 0.50

Agreement is weaker than it looks: the direct model also calls 44 of the 53 earlier candidates low. The controls make the run informative.

What the DFT run will settle

Reading the result

Controls first. The 4 Be/B crystals (predicted 25.3–103) must come out far above the 15, or the method is not separating low from high.

Then the 15. On PhoNIX (slide 20), picks as safe as these were truly ≤ 1 with precision 0.81 [0.68, 0.93] (451 crystals with MLIP γ , step 89). Carried over, that is about **12 of 15** (10–14), not 15 of 15. Unseen GNoME crystals: an expectation, not a promise.

Then the routes. Each DFT value is scored against stress, Poisson and direct – all three are already in `index.csv`.

Setup notes sent with the list

Six slanted cells (#5, 7, 9, 10, 11, 13): standardise before relaxing.

Spin-polarise Cr, Mn, Ni, Os(V), Ru(VI); check #4, #8, #15.

Heavy I, Te, Pt, Os, Au, Hf, Ta: spin-orbit per group practice.

MLIP soft modes to confirm: at most -0.077 THz.

Short of time: #2/#3 and #9/#10 are Hf/Zr twins – run one of each.

Reserves: use $\text{Cs}_2\text{CuAgO}_2$ and $\text{Fe}_3\text{SbCl}_7\text{O}$ first. The direct model doubts reserves 3–5 (2.2–2.6 W/m/K).

So far every number is model vs model or model vs database. This run tests the screen's picks against the real quantity.

SUMMARY

What this work shows

ALIGNN predicts the moduli better. One ALIGNN beats the three-model CGCNN ensemble on K (0.0549 vs 0.0630) and G (0.0726 vs 0.0781) – but not on the soft-end shear modulus, where the screen looks.

γ is the weak link. Against 45 measured κ_L , the paper's tabulated γ gives 0.226 (paper: 0.227); γ from Poisson's ratio gives 0.427.

Low calls hold up against phonon DFT. About 3 in 4 crystals the chain calls ≤ 1 really are (0.747, against 0.225 for a blind pick).

Every stored number is reproducible. 9,048 metric values rebuilt from their inputs, 0 mismatches.

Next: the DFT result. 15 crystals, 4 controls and 8 reserves are with the supervisor; we expect about 12 of 15 to land ≤ 1 .